

Trends' Signal Strength and the Performance of CTAs^{1,2,3}

Gert Elaut, Ghent University, BELGIUM

Péter Erdős, RPM Risk & Portfolio Management AB and Smooth Alpha Consulting,
SWEDEN and Artifex Holding Ltd., UK⁴

Abstract

We propose a new asset-based factor that relies on aggregating time series momentum signals over different horizons. Aggregating signals this way captures assets' trend signal strength, thereby addressing a limitation in existing time series momentum strategies. Our factor mimics a trend-following manager that increases exposure to markets where trends develop and decreases exposure to markets where trends fade. Taking into account a number of practical implementation issues, we find that our proposed factor performs better at replicating the stylized facts of CTAs' returns, allowing a more meaningful assessment of fund alpha.

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⁴*Corresponding author.*

1. Introduction

According to BarclayHedge, a hedge fund database with an extensive coverage of CTAs, total assets under management (AUM) in the CTA or Managed Futures industry stood at 333 billion USD at the end of the third quarter of 2015. This makes the CTA industry the second biggest hedge fund category after Fixed Income Arbitrage.

Until very recently, no asset-based benchmarks were available for the CTA industry. Instead, practitioners benchmarked CTAs' performance against manager-based indices. To some extent, the reliance on manager-based benchmarks is related to the challenge with constructing appropriate CTA benchmarks, as in these strategies there is no clear long or short bias. While CTAs generally disclose the list of markets traded, actual positions, which can be long or short, are unlikely to be revealed.

Benchmarking against peers has several limitations. First, manager-based benchmarks are essentially a combination of managers' alpha and beta. Second, [Fung and Hsieh \(2004\)](#) point out that hedge fund indices can be expected to inherit some of the biases that inherent in hedge fund databases. These effects make it hard to interpret the alphas estimated for individual managers as they may not accurately reflect managerial skill. Finally, manager-based indices tell us little about funds' style drift, sector exposures or other aspects of funds' performance that are typically assessed by investors.

Rather than benchmarking against peers, benchmarking managers against simple, transparent, and completely asset-based trend-following strategies may be more valuable. We contribute to the emerging strand of the literature building on the work of [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Baltas and Kosowski \(2013\)](#). First, we evaluate the performance of a trend-following strategy that captures signal strength by aggregating a wide range of short-term and longer-term signals. Whereas the original specification of [Moskowitz, Ooi, and Pedersen \(2012\)](#) relies on a binary signal, aggregating time series momentum signals of different lookback horizons results in a signal $S_t \in [-1, 1]$ that measures the strength of a trend. This allows the strategy to allocate to a position in proportion to a trend's strength. Such an approach is likely to be closer to actual practices of trend-following managers and

should yield a more robust time series momentum factor that better anticipates reversals. From a modeling perspective, aggregating over a wide range of potential specifications avoids an arbitrary choice of among candidate models. This considerably reduces data mining and calibration concerns. These model considerations are separate of the diversification benefits that justify combining signals over different horizons (see [Baltas and Kosowski \(2013\)](#), who show that time series strategies across different lookback horizons have low correlations).

Second, we incorporate a number of market frictions and real-life limitations, such as contract-specific transaction costs, the impact of exchange rate risk on contracts' profit-and-loss, and delays between signal generation and trade execution. Earlier work by [Hurst, Ooi, and Pedersen \(2013\)](#) points out the importance of some of these frictions. Incorporating these aspects ensures that our benchmark is an investable asset-based benchmark. Not taking into account real-life frictions tends to raise the bar for managers too much, hampering a meaningful interpretation of manager alpha.

We find that our strategy, referred to as adaptive time series momentum (ATSMOM), better matches a number of stylized facts of manager-based indices and outperforms existing benchmarks in explaining the returns of CTAs. Decomposing the ATSMOM factor using a principal component analysis, we find that another significant factor, which we coin the "speed factor", appears to be present in time series momentum's returns. This speed factor, which buys longer-term and sells shorter-term TSMOM strategies, is similar but not identical to the speed factor proposed by [Greyserman and Kaminski \(2014\)](#). We find that CTAs tend to load negatively on the speed factor. Fund alpha, however, is positively related to the exposure to the speed factor.

Turning to the performance evaluation using our new factors, we find that smaller CTAs exhibit a nearly even exposure to all asset classes under consideration. At the same time, larger funds tend to overweight more liquid (futures) markets, predominantly Fixed Income. While our asset-based factors capture much of the return variation of CTA managers, we find that some CTA managers outperform even on a risk-adjusted basis.

To investigate the drivers of the observed alpha vis-à-vis our model, we analyze the re-

relationship between risk-adjusted performance and fund characteristics. We find that fund characteristics only account for a small fraction of the cross-sectional variation in fund alphas. This suggests that the estimated alphas are indeed indicative of managerial skill. Interestingly, we also document persistence in the alphas, in that good annual performance in one year tends to repeat in the subsequent year. Finally, we find that contemporaneous fund flows do not affect the risk-adjusted performance of managers. This suggests capacity constraints are less of an issue for CTAs. This result echoes the findings of Baltas and Kosowski (2013), who rigorously test for capacity constraints in trend-following strategies and conclude that futures markets are liquid enough to accommodate the trading activity of the CTA industry.

2. A Stylized Example

Before we continue, we provide intuition of why combining different trend-following signals adds value and improves the overall performance of a strategy. Suppose we have two securities, whose price paths are reported in Table 1.

[Table 1 about here.]

We note that both securities have the same *initial* value and *terminal* value, and that the securities' returns exhibit identical levels of volatility over the period considered. In other words, both securities only differ in their interim price path. Application of a simple (long-term) time series momentum strategy over the period t through $t - 3$ yields a long signal in both instances. When we include the intermediate signals, however, we observe that the trends in both securities are considerably different. Aggregating all the time series momentum signals suggests that a reversal may be taking place in security A , whereas at t there is a strong and persistent trend in security B . This simple example suggests that aggregating signals over different lookback periods may add value as it captures trends' signal strength.

In addition to capturing signal strength, aggregating a wide range of individual signals has several other desirable properties as well. Obviously, the use of a smoother signal has

the advantage that it provides the manager with a signal that allows him or her to more gradually adjust positions to changing conditions. This particular property is distinct from a binary signal which generally dictates that any position should be reversed overnight.

To illustrate this feature, we present a time series momentum signal that captures signal strength (defined below) for futures data on the S&P 500 over the period 2007-2017. Figure 1 plots the result.

[Figure 1 about here.]

The result, reported in the bottom Panel of Figure 1 illustrates how a signal that incorporates the strength of the trend adjust more quickly to reversals. The signal, for example, is quicker in picking up the recovery in the second quarter of 2009. At the same time, in periods in which markets go sideways (e.g. 2015-2016), signal strength provides a smoother signal compared to the binary signal, which continuously moves from long to short.

3. Literature Review

Mutual funds are benchmarked against a combination of market indices and risk factors such as the factors suggested by [Fama and French \(1993\)](#) and [Carhart \(1997\)](#). Similarly, most hedge fund categories are evaluated against Fung and Hsieh's seven-factor (or eight-factor) model ([Fung and Hsieh, 2004](#)).⁵ While these factor models perform well in explaining the returns of mutual funds and most hedge fund categories, their performance in explaining CTAs' return variation is limited. Instead, the CTA industry still largely relies on manager-based indices.

Despite the apparent hurdles in modeling CTAs' strategies, there have been several studies that attempt to replicate their returns. [Fung and Hsieh \(2001\)](#) are among the first to focus on replicating trend-following hedge funds' returns. The authors suggest the use of primitive trend-following strategies (PTFS) based on lookback straddles, which capture the

⁵Or, more recently, against the factor model specification of [Buraschi, Kosowski, and Trojani \(2013\)](#), who propose a factor that tries to capture correlation risk.

returns of a market timer. Although implementing these factors in practice is possible, [Harvey, Rattray, Sinclair, and Van Hemert \(2016\)](#) note that this is neither straightforward nor cheap.

More recently, there has been renewed attention for modeling the returns accruing to CTAs. [Moskowitz, Ooi, and Pedersen \(2012\)](#) are the first to document, in a systematic manner, the presence of a 'trend' effect for a broad range of futures and forward contracts. They coin this effect time series momentum (TSMOM), which relies solely on the continuation of the price direction of the asset under consideration. This different from cross-sectional momentum, which relies on past winners outperforming past losers. [Moskowitz, Ooi, and Pedersen \(2012\)](#) show that a portfolio of TSMOM strategies, diversified across different asset classes, consistently delivers large and significant excess returns.

[Baltas and Kosowski \(2013\)](#) build on the work of [Moskowitz, Ooi, and Pedersen \(2012\)](#) to suggest a set of the Futures-Based Trend-Following Strategies. The authors extend the existing literature on time series momentum by considering weekly and daily strategies. [Baltas and Kosowski \(2013\)](#) also provide clear evidence that CTAs attempt to exploit momentum in the time series domain. In particular, the authors show that their suggested TSMOM factors significantly improve the explanatory power of multi-factor models applied to CTAs' returns and outperform the factors suggested by [Fung and Hsieh \(2001\)](#).

Our approach borrows from and extends the works of [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Baltas and Kosowski \(2013\)](#). In particular, we investigate the economic gains of using more than one or a limited number of signals. The use of a wide array of signals can be motivated along several lines. First, aggregating a large number of signals results in a combined signal that captures signal strength discussed in Section 2. This addresses a potential limitation in existing strategies that rely on a binary signal. It seems reasonable to assume that CTAs also consider the strength of a particular trend. Second, an investor ex-ante does not know the performance of a particular (e.g. a 15 day) specification. From this perspective a prudent approach is to combine a considerable number of signals. Relying on a limited number of signals based on historical performance also introduces some hindsight

bias. Such hindsight bias may raise the bar for managers too much, as pointed out by [Hurst, Ooi, and Pedersen \(2013\)](#). Combining different candidate signals, however, we avoid having to select a specific set of parameter specifications. This reduces model risk and at the same time enhances diversification.

Combining trend signals from different lookback periods matches a recent new avenue in academic research. In particular, [Han, Zhou, and Zhu \(2016\)](#) analyze the gains of combining signals from short-, intermediate, and long-term moving average signals in individual equities. They find that portfolios relying on a combination of price trends outperform portfolios formed based on a single price trend.

4. Data

4.1. Futures Data

The futures dataset that we use consists of the daily Close Price, Open Interest, and Volume for 98 futures contracts across four asset classes. Individual futures contract data are obtained from CSI and cover the period from January 1990 to September 2015. We report the list of futures contracts covered in [Table 2](#).

[Table 2 about here.]

Since futures contracts are short-lived contracts that expire at a predetermined date, we first construct a continuous time series of futures prices for each contract. In the online Appendix, we describe the particular approach used. The daily returns calculated from the continuous futures prices are equivalent to fully collateralized (unleveraged) returns in excess of the risk-free rate.⁶ As such, daily excess returns are constructed as

$$r_{i,t} = \frac{F_{i,t} - F_{i,t-1}}{F_{i,t}} \quad (1)$$

where F_t corresponds to the futures price of asset i at time t . The list of futures contracts that we employ is one of the most comprehensive used in the literature, as we include a

⁶For a thorough discussion, refer to Baltas and Kosowski (2015) and references therein.

number of metals-related futures and a number of currency pairs that are commonly traded by CTAs.

4.2. CTA Data

We collect monthly net-of-fee returns of live and dead funds labeled CTA in the BarclayHedge Database. We restrict the data on CTAs to the period from January 1994 to September 2015.⁷

We filter the sample of funds by looking at the self-declared strategy description and remove funds whose description is not consistent with the definition of a CTA. In the process, we identify the fund's flagship program and discard the other share classes. To account for backfill bias, we drop the first 12 observations (see Kosowski, Naik, and Teo, 2007).⁸ We also drop funds with (AUM) below 10 million USD to restrict the set of funds to the investable universe. Finally, we focus on funds that report their returns either in USD or EUR. The EUR-denominated returns and AUM are converted to USD.

We focus on systematic trend-following managers as their performance is most clearly related to time series momentum. These managers trade systematically and are diversified across many liquid futures contracts. Applying the above adjustments, we obtain a sample of 433 systematic trend-following CTAs. From this universe, we construct an AUM-weighted and an equal-risk weighted index, both rebalanced monthly.

⁷Although reporting to hedge fund databases is voluntary, Joenväärä, Kosowski, and Tolonen (2012) – in an analysis of the different publicly available hedge fund databases – conclude that BarclayHedge is the most comprehensive hedge fund database, especially for CTAs. Data from January 1994 is used to mitigate a potential survivorship bias, since most databases only started collecting information on defunct programs from 1994 onwards (see Joenväärä, Kosowski, and Tolonen, 2012)

⁸By keeping track of the number of months that are backfilled when a fund is first included in BarclayHedge database, RPM Risk & Portfolio Management AB tracked backfill bias in the BarclayHedge database for the period 2005-2010. For that sample period, the median (average) backfill bias was twelve (fourteen) months, consistent with the correction proposed by Kosowski, Naik, and Teo (2007).

5. Methodology

5.1. Adaptive Time Series Momentum

We construct a portfolio what we refer to as ATSMOM and which follows a strategy that is diversified both across time and across asset classes. The construction of our benchmark builds on the works of [Moskowitz, Ooi, and Pedersen \(2012\)](#) and [Baltas and Kosowski \(2013\)](#) (hereinafter MOP and BK, respectively). Analytically, using daily returns, a traditional diversified TSMOM strategy with a lookback of t days can be constructed as follows

$$r_{T+1}^{TSMOM} = \frac{1}{L} \sum_{l=1}^L \text{sgn}(r_{T-261, T-1}^l) \cdot \frac{0.4/\sqrt{261}}{\sigma_{T-60, T-1}^l} \cdot r_{T+1}^l \quad (2)$$

where sgn is the signum function, that is, $\text{sgn}(r_{T-t, T-1}^l)$ is the sign of the two-days lagged return over the lookback period $[T-t, \dots, T-1]$, L is the number of assets in the strategy and $\sigma_{T-60, T-1}^l$ is the two-day lagged exponentially weighted moving average (EWMA) estimator of volatility with a 60-day rolling window. Algebraically, the EWMA estimator in Eq. 2 is calculated as follows

$$\sigma_{T-60, T-1}^2 = (1 - \lambda) \cdot \sum_{t=0}^{59} \lambda^t \cdot (r_{T-t-1} - \bar{r})^2 \quad (3)$$

where λ is the decay factor, which we choose such that the center of mass is at around 60 days. We follow MOP in using this simple model for estimating volatility. The correction factor of 0.4 to the estimated volatility in Eq. (2) is suggested by MOP as to achieve an ex ante volatility of 40% per security. This is motivated from the observation that a 40% scaling factor can be expected to yield risk factors with an ex post volatility of around 12% per annum, which roughly matches the volatility of the equity risk factors of [Fama and French \(1993\)](#) (see [Moskowitz, Ooi, and Pedersen, 2012](#)). A typical CTA also targets around 12% volatility.

The ATSMOM strategy is defined as a time series momentum strategy whereby we average the signal for any given security in the portfolio over a wide set of lookback horizons

$$r_{T+1}^{ATSMOM} = \frac{1}{L} \sum_{l=1}^L \left(\frac{\sum_{t=10}^{260} \text{sgn}(r_{T-t, T-1}^l)}{251} \right) \cdot \frac{0.4/\sqrt{261}}{\sigma_{T-60, T-1}^l} \cdot r_{T+1}^l \quad (4)$$

We do not consider lookback periods of strictly less than 10 trading days. Trading using such short intervals is likely based on other methods than those based on daily closing prices (i.e. order flow and intraday data).⁹

From eq. (4) it is clear that the signal for every futures contract will vary between minus one and plus one (i.e. $S_t \in [-1, 1]$) depending on the strength of the trend. This is a desirable characteristics as a simple TSMOM strategy based on one lookback period can be criticized on the fact that it is a binary signal. As a result, a standard time-series momentum signal does not capture signal strength.¹⁰ As we illustrate in the introduction, our approach will allocate more to the futures contracts that exhibit clearer trends. When trends start to fade, however, the short-term signals will force the strategy to lower exposure more quickly than in the case of a strategy that only considers one long-term signal. At the portfolio-level, the strategy reduces exposure to markets where trends become less pronounced and adds to futures contracts where trends become more pronounced, in a more 'adaptive' way than a standard TSMOM strategy based on a single lookback horizon.

5.2. Incorporating Trading Frictions

In addition to improving the signal of existing TSMOM strategies, we also consider trading frictions. Incorporating such frictions may improve the explanatory power of our strategy as a benchmark for CTAs. First, the available benchmarks imply signal generation and trade execution on the same day, that is, for example, signal generation at the close price and entering the market during the same closing session. When re-balancing frequency is low, such as in the case of Moskowitz, Ooi, and Pedersen (2012) who employ monthly re-balancing, the impact of the exact closing price employed may be limited. In our case,

⁹Results for a trading strategy that also includes horizons between 1 and 9 days are qualitatively unchanged and are available up on request.

¹⁰In recent related works Lempérière, Deremble, Seager, Potters, and Bouchaud (2014) and Dudler, Gmür, and Malamud (2015) also suggest different TSMOM specifications that account for the signal strength.

however, the impact may be sizable as we re-balance and thus may shift positions daily.

In line with the work of [Hurst, Ooi, and Pedersen \(2013\)](#), we systematically skip one trading day between signal generation and trade execution. For example, we only enter a position at Tuesday's closing price if that decision relies on a signal generated based on Monday's closing price. Similarly, the first day we can close that same position is during Wednesday's closing session and the return of such a position will be the percentage price difference between Wednesday's and Tuesday's closing prices.

We also consider the impact of the exchange rates in cases where the futures are traded in a currency other than USD. We assume that the collateral or margin is always held in the base currency (USD). Thus, *only* the daily profit and loss (P&L) generated from positions in the contracts traded in a foreign currency is exchanged to USD at the daily closing exchange rate. The margin itself, which is held in domestic currency, is not exposed to exchange rate risk (see Appendix A in [Koijen, Moskowitz, Pedersen, and Vrugt, 2016](#)).

To allow for a meaningful performance measurement, we account for transaction costs. A prerequisite to incorporate costs is, of course, appropriate estimates of the trading costs actually incurred by CTAs. To this end, we estimate the explicit trading costs from actual charges incurred by the managers in one of RPM Risk & Portfolio Management's¹¹ flagship funds over a one-year period from August 2013 through August 2014. Explicit trading costs include gross commissions, clearing fees, exchange fees, NFA (National Futures Association) charges, and brokerage and execution charges.

Second, we also account for implicit transaction costs arising from the bid-ask spread that traders usually pay market makers for providing liquidity.¹² Using the bid-ask spread to estimate implicit transaction costs dates back to [Demsetz \(1968\)](#), who argues that when participants trade through market makers, they will pay the difference between the true

¹¹RPM Risk & Portfolio Management AB is a fund-of-funds specializing in CTA strategies and liquid Global Macro managers, based in Stockholm (Sweden).

¹²Effective spread estimators ([Roll, 1984](#); [Smith and Whaley, 1994](#)) and approaches to estimate bid-ask spread directly from the order book ([Locke and Venkatesh, 1997](#)) have also been proposed. [Szakmary, Shen, and Sharma \(2010\)](#) and [Locke and Venkatesh \(1997\)](#) point out, however, that these estimates are close to the tick size. Since estimating the bid-ask spread from the order book is beyond the scope of the current paper, we stick to the simplification that the tick size is a good proxy for the bid-ask spread.

price and the bid or ask price on every trade. Following the standard approach in the literature, in a round trip, we approximate the bid-ask spread by the reported tick-size for every contract. For completeness, we note that transaction costs are likely to be a nonlinear function of trading volume. To estimate the relationship between volume and transaction costs requires the price at which a trade arrives in the market as well as trading volume. We refer to the work of [Frazzini, Israel, and Moskowitz \(2012\)](#) for more details for estimating nonlinear transaction cost functions and their implications for exploiting asset pricing anomalies. Taking into account transaction costs and other frictions (such as position limits) requires assumptions on the strategy's size. We refer to the work of [Greyserman and Kaminski \(2014\)](#) who analyze the implications of fund size on trend-following strategies' implementability. In this study, we assume that transaction costs increase linearly with trading volume.

Unfortunately, we only have data for transaction costs in recent years. Following [Hurst, Ooi, and Pedersen \(2012\)](#), we assume that in the first half of the sample period (1991-2002), transaction costs were twice as high as in the second half of the sample period (2003-2015).¹³

We find that trading costs vary considerably over the different asset classes; short-rate futures are the least expensive to trade, though these contracts are also the least volatile. Trading in VIX and grains futures is most expensive. This finding is mainly driven by large tick size indicating lower liquidity in these markets. On average, across all markets traded, we find that the bid-ask spread is responsible for almost three quarters of the overall transaction costs.

¹³When a futures contract is rolled over to a further-dated contract, the strategy closes the nearby contract and opens a position in the new contract. The date of the contract rollover coincides with the rollover used for the construction of the continuous futures (see online Appendix for the continuous contract construction as well as the estimated transaction costs). On such days, turnover is usually much higher than on other days. Daily turnover of the strategy fairly limited, except in the case of short rate futures. These contracts exhibit very low levels of volatility (0.01% average daily volatility) compared to other contracts (1.2% average daily volatility) and thus require a large notional position to obtain similar levels of volatility. Each short rate futures included generates an average daily turnover of around 22-23%, whereas, the average turnover for the other contracts is just 0.3%.

6. Results

6.1. ATSMOM's Stand-Alone Performance

Table 3 reports performance statistics for our strategy as well as for the factors suggested by Moskowitz, Ooi, and Pedersen (2012) and Baltas and Kosowski (2013).¹⁴ ATSMOM is reported both gross and net of transaction costs in Panel A. The other benchmarks, in Panel B, are gross of transaction costs. All the factors are scaled to 10% ex-post volatility.

[Table 3 about here.]

We observe that ATSMOM yields somewhat higher minimum and maximum returns than MOP and the BK_D and BK_W . This suggests that the strategy is successful at limiting downside risk and allocating more to better performing assets. The lower downside risk is likely the consequence of signal diversification benefits as well as the higher re-balancing frequency. More frequent re-balancing alone, however, does not guarantee lower downside risk, as is evident from BK_D 's maximum drawdown (MDD). Taking into account transaction costs, the benefits resulting from the more pro-active nature of ATSMOM clearly come at a cost; the Sharpe ratio net-of-transaction costs drops to 0.96.

The higher upside of ATSMOM also translates to a higher skewness. Positive skewness is consistent with one of the stylized facts of CTAs. These funds tend to produce positively skewed returns (refer to, among others, Fung and Hsieh (2001), Lamm Jr (2005), and Ding and Shawky (2007)). This feature is also present in the BK_W and BK_D . Before transaction costs, ATSMOM reports slightly lower average annual returns than MOP. The difference in risk-adjusted performance is, however, not statistically significant (using the approach of Ledoit and Wolf (2008) to test the statistical significance of the difference in Sharpe ratios, we obtain a p -value of 0.288).

Although the focus of this work is not on stand-alone performance, the results so far indicate that our proposed benchmark is able to compete with existing benchmarks. In

¹⁴The time series momentum factor (henceforth MOP) is available from AQR's website. The monthly, weekly, and daily Futures-based Trend-following Benchmarks (FTB, henceforth BK_M , BK_W , and BK_D) are available from Robert Kosowski's website.

addition, transaction costs do not erode risk-adjusted performance. Next, we turn to the use of ATSMOM as a benchmark for the CTA industry. Does an approach that considers signal strength better match CTAs' returns?

6.2. A Benchmark for CTAs

Panel C of Table 3 reports the performance of existing industry indices. These indices are often used by practitioners to benchmark individual managers. The two most commonly used benchmarks are the BarclayHedge CTA Index and the SG CTA Index. BarclayHedge also publishes a large cap index called BTOP 50 and SG a Trend-Following sub-index.¹⁵ In addition to these manager-based indices, SG also constructs an asset-based benchmark called the SG Trend Indicator index which reflects the returns of a strategy that relies on a simple 20/120 moving average crossover model. The index is reported net of transaction costs and a hypothetical 2% management and 20% performance fee.

In addition to the above indices, we construct an AUM weighted and an equal risk-weighted (ERW) index using the systematic trend-followers from Section 4. These indices too are not investable as one cannot re-balance a CTA portfolio on a monthly basis. Lengthy due diligence and legal processes to opening and closing managers makes such an approach impractical. Nevertheless, the indices are representative of then-current CTAs.

Most of the indices exhibit positive skewness, drawdowns of approximately 15% at 10% annual volatility (with the exception of the BTOP 50), and Sharpe ratios between 0.31 and 0.93. The Trend Indicator reports the highest drawdown, perhaps the result of the fact that the strategy employs just one medium-term moving average crossover.

Figure 2 plots the 3-year rolling window Sharpe ratio of the different benchmarks reported in Panel A and B of Table 3.

[Figure 2 about here.]

¹⁵The BarclayHedge and SG manager-based indices are equal-weighted. As a consequence, these indices are overweight CTAs that target higher levels of volatility. The manager-based indices are rebalanced once a year. The BarclayHedge CTA index is a broad index of CTAs, some of which are not trend-followers nor systematic. The SG CTA index includes only the largest 20 CTAs that are open to investment and report performance and AUM on a daily basis.

The performance of ATSMOM is almost always between the slower(-to-react) MOP and faster BK strategies and is less likely to significantly out or under-perform the other benchmarks. Longer-term strategies usually outperform shorter-term strategies. This was clearly the case during 2013 through 2015, when MOP clearly outperformed ATSMOM. However, in periods when shorter-term strategies outperform, longer-term strategies tend to suffer. Greyserman and Kaminski (2014) note that it may be difficult, if not impossible, to determine ex-ante the horizon that will perform best over a given period. In such an environment, trading based on a wide set of signals may be preferable.

Moving beyond summary statistics, we investigate the relationship between our proposed ATSMOM strategy and existing (equity-based) risk factors, the primitive trend-following strategies (PTFS) of Fung and Hsieh (2001) and a number of other recently proposed risk factors more formally in Table 4.

[Table 4 about here.]

Column (1) regresses the returns of ATSMOM on a number of equity-based risk-factor. In particular, we include the factors of Fama and French (1993), the Carhart (1997) cross-sectional momentum factor and the tradeable (equity-based) liquidity factor of Pástor and Stambaugh (2003). We find that only cross-sectional momentum is related to our benchmark. Next, we consider the results for 8-factor model of Fung and Hsieh (2004) in column (2), where we include all five PTFS factors rather than just the commodity, bond, and foreign exchange PTFS. Whereas the extended FH model generally works well for most hedge fund categories (see Fung and Hsieh, 2004), only the PTFS factors are significant in explaining ATSMOM.

We follow the work of Moskowitz, Ooi, and Pedersen (2012), Asness, Moskowitz, and Pedersen (2013), and Koijen, Moskowitz, Pedersen, and Vrugt (2016) and regress ATSMOM's returns on a number of macroeconomic, liquidity, volatility, and sentiment variables. The results, reported in the online Appendix, show that variation in these variables does not explain ATSMOM's excess returns.

We also regress the strategy's returns against the Global Value and Global (cross-sectional) Momentum factors proposed by [Asness, Moskowitz, and Pedersen \(2013\)](#), which are more appropriate since these factors cover multiple asset classes. The results, reported in column (3), indicate that these factors perform somewhat better in explaining the variation in the strategy's returns, with both coefficients being significantly positive. Our strategy, however, continues to generate a significant and substantial alpha of 5.64% p.a. *vis-à-vis* these factors.

Table 5, Panel A, reports the explanatory power of a number of asset-based style regressions, where we regress the manager-based CTA indices against the competing asset-based benchmarks. We consider the period from January 2000 through January 2012, for which data for all benchmarks is publicly available.

[Table 5 about here.]

Fung and Hsieh's PTFS explain up to 30% of the variation in the CTA indices. The 10-factor model, which considers other hedge fund asset-based style factors in addition to the PTFS, performs only marginally better. Turning to the SG Trend Indicator, an industry benchmark that has gained some traction among practitioners in the CTA industry, we find that this indicator performs surprisingly well over the sample period. MOP also performs consistently across the CTA benchmarks and produces R²s of around 45%, slightly lower than that of the Trend Indicator. The three-factor model of [Baltas and Kosowski \(2013\)](#) yields comparable results, with adjusted R²s ranging from 40% to 50%. ATSMOM, however, performs best across the board.

To determine whether the observed increase of our proposed factor in capturing CTAs' returns is meaningful, we compare ATSMOM to the model proposed by [Baltas and Kosowski \(2013\)](#).¹⁶ To this end, we first estimate the incremental value added from using ATSMOM by calculating the residuals from a regression that regresses ATSMOM against the Futures-based Trend-following Benchmark (FTB) Strategies. For comparison purposes, we scale all the regressors including the residuals to 10%. We then rerun the specification of [Baltas](#)

¹⁶We refrain from using an incremental *F*-test because of potential multicollinearity issues. The pairwise Pearson correlation between our factor and the average of BK's factors is 0.8.

and Kosowski (2013), including the obtained residuals. If the coefficient on the residuals is statistically significant, then this confirms that our proposed factors adds value over and above the FTBs. The results are reported in Panel B of Table 5.

Not only do we find that the coefficient is significant at conventional levels, and leads to a meaningful increase in the explanatory power of the models (i.e. a 15 to 20 percentage points increase compared to the initially reported adjusted R-squared, see Panel A), we also observe that the relationship is economically significant. In particular, scaled to the same volatility, we find that the coefficient on the residuals is comparable in magnitude to Baltas and Kosowski's monthly and weekly factor. This indicates that the BK factors alone may fall short of explaining CTAs' returns.

6.3. Decomposing Adaptive Time Series Momentum

Our proposed strategy is equivalent to an equal-weighted combination of a set of TSMOM portfolios using lookback horizons from 10 days to 260 days. ATSMOM discussed so far trades the net position. We can look at these 251 portfolios as separate variables, jointly describing trend-following strategies. In an attempt to better understand CTAs' returns, we decompose the ATSMOM's returns into its constituent (significant) factors. The question we wish to answer is whether a single factor, which we call ATSMOM and which is a simple average of a wide set of TSMOM portfolios, is enough to fully describe time series momentum strategies in general. The evidence in Greyserman and Kaminski (2014) suggests that there may be other factors beyond ATSMOM driving CTA returns.

One way to address this empirical question is to employ a principal component analysis (PCA) to the constituting TSMOM portfolios. To analyze the statistical significance of the different principal components in time series momentum returns, we draw 10,000 bootstrapped samples (see Peres-Neto, Jackson, and Somers, 2003) to calculate p -values for the estimated eigenvalues. The eigenvalues are compared to both the broken-stick and Marcenko and Pastur distribution (see e.g., Süß, 2012).

At the 90% confidence level, both distributions indicate that the first three principal components, corresponding to the three largest eigenvalues, are significant. At the 95% level of

significance, the Marcenko-Pastur critical values point towards three significant components. The broken-stick model, however, suggests only the first two PCs are significant. Regressing the manager-based indices against the first three PCs, we find that only the first two are significant.

The results indicate that the first principal component (PC1) is similar to an equal-weighted portfolio of TSMOM portfolios, which is consistent with the definition of ATSMOM.¹⁷ Indeed, the first PC shows a correlation of 0.99 with ATSMOM's net returns. ATSMOM, by design, assigns an equal weight to each TSMOM strategy with a lookback window between 10 and 260 days. This implies that there is a significant amount of overlap in the lookback windows. For example, the 10-day window is also part of the 11-day, the 12-day, up to the 260-day window (though it becomes increasingly less important in determining the trend). To generalize, any N-day window is part of all longer-term windows. This means that the short-term windows are generally "over-weight" in the overall strategy. The results corroborate this conjecture, with both PC1 and PC2 collectively loading more on the shorter-term signals. These dynamics may explain the large significance of the first two eigenvalues.

The second principal component (PC2), however, does not load uniformly on the different constituent portfolios. Instead, we find that PC2 is equivalent to a strategy that buys shorter horizon (strategies that react fast to changes in trends) and sells longer horizon momentum strategies (strategies that react slowly to changes in trends). It can therefore be interpreted as a "speed factor". The factor is close to the opposite of the speed factor in [Greyserman and Kaminski \(2014\)](#), which buys longer (slower) and sells shorter-horizon (faster) momentum strategies. Nevertheless, without loss of generality, we take the negative of PC2 to get a speed factor similar to that of [Greyserman and Kaminski \(2014\)](#). Principal components are indifferent to scaling since they are extracted in a way to show zero pairwise correlation.

One possible practitioner criticism for this approach relates to correlation. When equal weighting the signs of a set of returns that are overlapping the signals are not independent. This means that the ATSMOM factor is loaded heavily on the longer time horizons than

¹⁷Results available in the online Appendix.

the shorter ones. In this case, if managers in the space account for this when measuring and combining different signals this could make ATSMOM a longer term tilted signal by construction.¹⁸

We know that longer-term momentum strategies tend outperform their shorter-term counterparts. At the same time, longer-term strategies also exhibit lower skewness (see Table 3). Return to the speed factor may therefore be a compensation for bearing tail risk that results from longer-term strategies (for a more detailed discussion, see [Greyserman and Kaminski \(2014\)](#)).

Next, we investigate the Sharpe ratio of a portfolio that combines ATSMOM with the speed factor, net of transaction costs, as a function of the weight of the speed factor. If the speed factor is scaled to the volatility of ATSMOM then, through diversification and lower trading costs, we find that the speed factor contributes positively to the overall performance if its weight is capped at 20%. Diversification follows from the fact that, by construction, the speed factor has a correlation of zero with ATSMOM.

We note that, although ATSMOM is a tradable factor, PC2 is not yet tradable. This is because it is based on loadings that are estimated in-sample. In addition, the factor is not net of transaction costs. To turn it into a tradable factor, we calculate the weights of the horizon portfolios in the speed factor at any point in time. The weights are proportional to the loadings estimated using an expanding window up to the penultimate day, to avoid a look-ahead bias. The initial estimation period is one year.

While we do not argue that a CTA would trade our proposed speed factor on a stand-alone basis, the factor can be used as an overlay to complement a more general trend-following strategy, in which case only the net position is traded. From this perspective, only the additional trading costs related to the speed factor are relevant. In what follows, we discuss the speed factor's performance from this perspective.

While by construction unrelated to ATSMOM, we find in unreported results that the speed factor is related to BK's factors, PFTS, cross-sectional momentum, and liquidity.

¹⁸We thank an anonymous referee for this valuable insight.

The (positive) loading on liquidity may be surprising in light of our earlier finding that the ATSMOM is unrelated to liquidity risk. The speed factor, however, invests in longer-term (slower-to-react) momentum and sells shorter-term (faster-to-react) momentum strategies and can thus be expected to be more exposed to liquidity risk as it is expected to react only slowly and may be more likely to trade during than ahead of periods of distress and lower liquidity.

6.4. *The Speed Factor, Asset Class-Based Factors, and CTA Performance*

With the introduction of the speed factor, we repeat the previous analysis where we regress the various CTAs indices against the newly introduced factors. We also extend the analysis by considering asset-class specific ATSMOM for commodity, equity, fixed income, and FX futures. The asset class-based factors are scaled to 10% volatility. Table 6 reports the results.

[Table 6 about here.]

As we have already discussed in the previous section, ATSMOM is able to explain a substantial part of the variation in CTAs' returns (Table 5). This suggests that ATSMOM captures CTAs' trading behavior fairly accurately. Extending the model with the speed factor increases the fit of most of the regressions, with the exception of the SG indices (Table 6, Panel A).

In Panel B of Table 6, we report the results for the asset class-based ATSMOM. Applying asset class-based ATSMOM has two advantages over a diversified ATSMOM strategy. First, the asset class-based benchmarks improve the explanatory fit since we leave the relative importance of the different sectors unconstrained. Second, asset class benchmarks allow for a style analysis. Since we have scaled the asset class-based factors to 10% volatility p.a., the loadings can be compared directly. Table 6, Panel B, indicates that CTAs allocate most to fixed income futures and least to FX and commodity futures. The weight of the asset class, however, tends to depend on fund size; large capitalization indices – the BTOP50 and the AUM-weighted Barclay systematic TF – invest more in more liquid markets, i.e.,

fixed income and less in commodities. Small capitalization managers, proxied by the Barclay CTA and equal risk-weighted Barclay systematic TF indices, invest more evenly across asset classes.

Next, we turn to individual CTAs. We apply the asset-class based ATSMOM and the speed factor to all the individual funds included in the BarclayHedge sample with at least a one-year track record after inclusion (see Section 4).¹⁹ Table 7 reports the mean and median of the parameter estimates for 335 funds that have produced jointly significant betas at the 10% level of significance.

[Table 7 about here.]

On average, our model is able to explain 40% of the variation in individual CTAs' returns. The average (median) alpha is positive at 0.29% (0.82%) p.a., with 16% of the fund alphas significantly positive and 6% significantly negative. For the funds for which we obtain a significant alpha, we observe considerable variation; funds with positive alphas generate mean (median) alphas of 4.77% (3.91%) p.a. whereas funds with negative alphas underperform the ATSMOM strategies by an average (median) of 9.55% (6.56%) p.a.

Interestingly, Fixed Income is significant for 70% of the funds. Thus, CTAs tend to be exposed to Fixed Income most frequently and this result corroborates the earlier finding that CTA indices load most heavily on this factor. Commodities is the second most important one, being significant in 64% of the cases. Equity is significant in 53% and FX is significant for 48% of the funds. The speed factor is also an important driver of CTA returns being significant in half of the regressions. The negative average coefficient of the speed factor is an indication that managers tend to add shorter horizons momentum strategies as a layover to their main strategy. This result is the consequence of ATSMOM being a longer term tilted strategy accounting for correlation between the constituting signals. As managers incorporate this aspect in their signal weights, they have more emphasis on shorter horizons.²⁰

¹⁹We note that dropping funds that stop reporting before turning two years unavoidably introduces some survivorship bias.

²⁰We thank an anonymous referee for this valuable insight.

Having obtained individual alphas for the CTAs, we investigate the role of fund characteristics in generating alpha. We regress the alpha for each fund for every year on yearly fund characteristics that include lagged alpha, fund size, fund age, standard fund flow variable, R^2 and the relative factor exposures of the performance regressions, the level of management and incentive fees, margin-to-equity (ME) ratio, and round turns per million dollars per year.

The estimated alphas are subject to measurement error. If we do not account for this, the measurement error will generate heteroscedasticity in the panel regression residuals making standard significance tests invalid. To correct this, we weigh each observation by the reciprocal of the standard errors of the performance regressions, as in [Dahlquist, Engström, and Söderlind \(2000\)](#).

Table 8 reports the results controlling for time-fixed effects. Column (1) omits margin-to-equity (ME) ratio and round turns per million USD per year statistics, as these are only available for a subset of the CTAs. Specifications that includes the ME ratio and round turns per million are reported in column (2) and (3) of Table 8, respectively.

[Table 8 about here.]

The results in Table 8 suggest persistence in CTAs' performance. CTAs that outperformed our benchmarks over the previous year tend to repeat that superior performance the following year. Fund size negatively affects risk-adjusted performance. Somewhat surprising though, age is positively related to alpha. However, for instance, the expected risk-adjusted performance of a five year old CTA that has 1 billion USD under management is, *ceteris paribus*, 1.7% p.a. less than that of a CTA that manages only 10 million USD but it is only two years old. This indicates that interpreting one of the variables alone can be misleading.

Contemporaneous fund flows do not affect risk-adjusted performance, suggesting that capacity constraints are generally not an issue. Adding the R^2 s of the performance regressions, we test and reject the hypothesis in [Sun, Wang, and Zheng \(2012\)](#) that hedge funds whose returns are less explainable by risk factors bear more managerial skills. In contrast, funds

that engage in pure trend-following approaches tend to generate higher risk-adjusted performance. Thus, alpha does not appear to derive from being less mainstream, but from other sources. This may include superior risk management and better trade execution leading to lower implicit and explicit transaction costs.

The relative factor weights are simply calculated from absolute loadings in the individual performance regressions. All else equal, we find that higher equity momentum exposure is likely to result in higher risk-adjusted performance. In contrast, funds with higher allocations to Fixed Income TF strategies tend to generate lower alpha. Interestingly, CTAs that have higher exposure to the speed factor significantly outperform those who have less exposure. The speed factor exposure may proxy for the level of sophistication of the manager, since our results suggest that there is some benefit from allocating to the speed factor in terms of diversification and lower transaction costs.

Higher margin usage over capital invested (ME ratio) appears to be a sign of better performance, most probably through economies of scale. This result suggests that increased risk-taking does not, per se, imply inferior risk management and poorer performance. Finally, more trading in terms of rounds per million USD per year does not affect risk-adjusted performance. Only a small part of the cross-sectional variation in estimated alphas is attributable to fund characteristics such as past performance, fund age, fund size, fees, and style. We conclude that the alphas obtained vis-à-vis our new risk factors can, to some extent, be interpreted as capturing managerial skill.

7. Conclusion

In this paper we propose a novel trend-following strategy benchmark that incorporates signal strength and relies on an adaptive time series momentum (ATSMOM) strategy opposed to the simpler TSMOM strategy that builds on a binary long/short signal. ATSMOM changes exposure to futures markets dynamically by aggregating time series momentum signals over a wide range of horizons. This way, the model increases the allocation to the markets where trends are more well-behaved and decreases exposure to markets where trends

are fading.

Decomposing time series momentum strategies' returns, we find that, along with ATSMOM, a second factor is present which we coin the speed factor as it buys longer and sells shorter term time series momentum strategies. In addition, we also incorporate a number of real-life frictions that may impact strategy returns.

We find that ATSMOM alone or augmented with the speed factor better explains Managed Futures funds' reported returns. As such, our approach can aid practitioners in benchmarking and manager selection. We document that a subset of funds continues to exhibit positive alpha compared to our new risk factors. Moreover, the abnormal returns of these funds can only be partly explained by observable fund characteristics and thus appear indicative of skill.

Importantly, we document strong momentum in CTA risk-adjusted performance, as strong performance in one year tends to repeat in the subsequent year, and find evidence that fund size is negatively, whereas fund age is positively related to risk-adjusted performance. Fund style, i.e., asset class exposure and the applied trading strategy, also contributes to CTA alphas. Allocation to the speed factor also appears to be related to manager skill. Contemporaneous fund flows, in contrast, do not affect risk-adjusted performance, suggesting capacity constraints are less an issue for CTAs. Higher management and performance fees do not signal prospect for better performance.

Finally, we note that while our analysis captures many of the real-life frictions associated with systematic trading, it does not consider the nonlinear relationship between trading volume and market impact. Larger funds may be forced to trade more liquid markets as well as relatively slower-moving signals (see [Greyserman and Kaminski \(2014\)](#)). This implies that our factor may still be setting the bar too high for funds with large assets-under-management who tend to be more constrained in this regard.

References

- Asness, C. S., T. J. Moskowitz, L. H. Pedersen, 2013. Value and Momentum Everywhere. *The Journal of Finance* 68(3), 929–985.
- Baltas, N., R. Kosowski, 2013. Momentum Strategies in Futures Markets and Trend-following Funds. .
- Buraschi, A., R. Kosowski, F. Trojani, 2013. When there is no place to hide: Correlation risk and the cross-section of hedge fund returns. *The Review of Financial Studies* 27(2), 581–616.
- Carhart, M. M., 1997. On Persistence in Mutual Fund Performance. *The Journal of Finance* 52(1), 57–82.
- Dahlquist, M., S. Engström, P. Söderlind, 2000. Performance and Characteristics of Swedish Mutual Funds. *Journal of Financial and Quantitative Analysis* 35(3), 409–423.
- Demsetz, H., 1968. The Cost of Transacting. *The Quarterly Journal of Economics* 82(1), 33–53.
- Ding, B., H. A. Shawky, 2007. The Performance of Hedge Fund Strategies and the Asymmetry of Return Distributions. *European Financial Management* 13(2), 309–331.
- Dudler, M., B. Gmür, S. Malamud, 2015. Momentum and Risk Adjustment. *The Journal of Alternative Investments* 18(2), 91–103.
- Elaut, G., M. Frömmel, J. Sjödin, 2015. Crystallization: A Hidden Dimension of CTA Fees. *Financial Analysts Journal* 71(4), 51–62.
- Fama, E. F., K. R. French, 1993. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33(1), 3 – 56.
- Frazzini, A., R. Israel, T. J. Moskowitz, 2012. Trading costs of asset pricing anomalies. .

- Fung, W., D. A. Hsieh, 2001. The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers. *Review of Financial Studies* 14(2), 313–341.
- Fung, W., D. A. Hsieh, 2004. Hedge fund benchmarks: A risk-based approach. *Financial Analysts Journal* 60(5), 65–80.
- Greyserman, A., K. Kaminski, 2014. *Trend Following with Managed Futures: The Search for Crisis Alpha*. John Wiley & Sons, .
- Han, Y., G. Zhou, Y. Zhu, 2016. A trend factor: Any economic gains from using information over investment horizons?. *Journal of Financial Economics* 122(2), 352 – 375.
- Harvey, C. R., S. Rattray, A. C. Sinclair, O. Van Hemert, 2016. *Man vs. Machine: Comparing Discretionary and Systematic Hedge Fund Performance*. .
- Hurst, B., Y. H. Ooi, L. H. Pedersen, 2012. A century of evidence on trend-following investing. AQR white paper.
- Hurst, B., Y. H. Ooi, L. H. Pedersen, 2013. Demystifying managed futures. *Journal of Investment Management* 11(3), 42–58.
- Joenväärä, J., R. Kosowski, P. Tolonen, 2012. *Revisiting Stylized Facts about Hedge Funds*. Imperial College Business School.
- Koijen, R., T. Moskowitz, L. Pedersen, E. Vrugt, 2016. Carry. Fama-Miller Working Paper. Available at SSRN 2298565.
- Kosowski, R., N. Y. Naik, M. Teo, 2007. Do hedge funds deliver alpha? A Bayesian and bootstrap analysis. *Journal of Financial Economics* 84(1), 229 – 264.
- Lamm Jr, R. M., 2005. The answer to your dreams? Investment implications of positive asymmetry in CTA returns. *The Journal of Alternative Investments* 7(4), 22–32.
- Ledoit, O., M. Wolf, 2008. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance* 15(5), 850 – 859.

- Lempérière, Y., C. Deremble, P. Seager, M. Potters, J.-P. Bouchaud, 2014. Two centuries of trend following. *Journal of Investment Strategies* 3(3), 41–61.
- Locke, P. R., P. C. Venkatesh, 1997. Futures market transaction costs. *Journal of Futures Markets* 17(2), 229–245.
- Moskowitz, T. J., Y. H. Ooi, L. H. Pedersen, 2012. Time series momentum. *Journal of Financial Economics* 104(2), 228 – 250 Special Issue on Investor Sentiment.
- Pástor, L., R. F. Stambaugh, 2003. Liquidity Risk and Expected Stock Returns. *Journal of Political Economy* 111(3), 642–685.
- Peres-Neto, P. R., D. A. Jackson, K. M. Somers, 2003. Giving Meaningful Interpretation to Ordination Axes: Assessing Loading Significance in Principal Component Analysis. *Ecology* 84(9), 2347–2363.
- Roll, R., 1984. A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market. *The Journal of Finance* 39(4), 1127–1139.
- Smith, T., R. E. Whaley, 1994. Estimating the effective BID/ASK spread from time and sales data. *Journal of Futures Markets* 14(4), 437–455.
- Sun, Z., A. Wang, L. Zheng, 2012. The Road Less Traveled: Strategy Distinctiveness and Hedge Fund Performance. *Review of Financial Studies* 25(1), 96–143.
- Süss, S., 2012. The pricing of idiosyncratic risk: evidence from the implied volatility distribution. *Financial Markets and Portfolio Management* 26(2), 247–267.
- Szakmary, A. C., Q. Shen, S. C. Sharma, 2010. Trend-following trading strategies in commodity futures: A re-examination. *Journal of Banking & Finance* 34(2), 409 – 426.

Table 1: Example Aggregating Trend-following Signals

	Security A	Security B	$r_{t,A}$	$r_{t,B}$
$t - 3$	90	90		
$t - 2$	130	83	44.44%	-8.28%
$t - 1$	140	120	7.69%	45.37%
t	125	125	-10.71%	4.17%
	Signal	Signal	σ_A	σ_B
$Sign(t - 1, t)$	-1	1	28.08%	28.08%
$Sign(t - 2, t)$	-1	1		
$Sign(t - 3, t)$	1	1		

Table 2: List of Futures

Foreign Exchange	Fixed Income	Equities	Commodities
USD/MXN	Australian 10-YR	CAC-40	Gas Oil
USD/SEK	Canada 10-YR	Nikkei 225	Natural Gas
USD/GBP	Eurodollar 3-M	Russell 2000	Brent Crude
USD/CAD	Australian 3-YR	S&P Midcap 400	Heating Oil
USD/JPY	Japan 10-YR	Hang Seng	Light Crude
USD/AUD	Gilt	DAX	Unleaded Gas
USD Index	US 90-Day Bank Bill	S&P 500	RBOB
EUR/USD	US 3-YR	TOPIX	Copper
USD/ZAR	US 2-YR	FTSE 100	Platinum
USD/BRL	US 5-YR	Swiss Market	Silver
USD/SEK	US 10-YR	INEX 35	Gold
USD/NOK	Muni Note	MIB 30	Palladium
USD/NZD	BUXL	NASDAQ 100	Live Cattle
AUD/NZD	BUND	MSCI Taiwan	Live Hogs
AUD/JPY	BOBL	DJIA	Pork Bellies
EUR/JPY	SCHATZ	KOSPI 200	Feeder Cattle
EUR/NOK	Korean 3-YR	DJ Stoxx 50	Corn
EUR/SEK	PIBOR	DJ Euro Stoxx	Oat
EUR/GBP	EURIBOR 3-M	S&P Canada 60	Soybeans
EUR/CHF		CBOE VIX	Soybean Meal
		OMX	Soybean Oil
		US MSCI EAFE	Wheat W
		AEX	Wheat
		NYSE Comp	Cocoa
		All Ordinary SPI	Cotton No. 2
		SPI 200	Coffee
			Fr. Orange Juice
			Sugar No. 11
			Lumber
			Nickel

Table 3: Descriptive Statistics for Asset- and Manager-based CTA Benchmarks

	Panel A - ATSMOM		Panel B - Existing Asset-Based Benchmarks			
	ATSMOM -		BK Monthly -		BK Daily -	
	Gross	ATSMOM - net of TCs	MOP - Gross	Gross	Gross	BK Average - Gross
Min (%)	-5.78	-5.91	-8.04	-7.11	-8.43	-7.81
Max (%)	15.26	15.22	10.50	8.88	15.29	12.38
Ann. Mean (%)	11.05	9.62	12.42	12.67	10.98	9.00
Ann. Median (%)	9.05	7.73	12.61	13.80	10.16	6.62
Ann. St. Dev. (%)	10.00	10.00	10.00	10.00	10.00	10.00
Skewness	0.70	0.71	0.04	-0.20	0.63	0.77
Kurtosis	5.55	5.61	3.26	3.12	5.85	4.81
Sharpe	1.11	0.96	1.24	1.27	1.10	0.90
Sortino	2.26	1.93	2.24	2.07	2.09	1.95
Max DD (%)	-11.24	-14.13	-13.23	-14.96	-10.17	-13.12
Period	01/'92-09/'15	01/'92-09/'15	01/'92-09/'15	01/'92-01/'12	01/'92-01/'12	01/'92-01/'12

	Panel C - Existing Industry Benchmarks			
	Barclay CTA	BTOP 50	SG CTA	SG Trend
	BarclayHedge Systematic TF	BTOP 50	SG CTA	SG Trend Indicator
Min (%)	-6.55	-8.30	-9.03	-6.28
Max (%)	8.86	10.86	9.66	11.09
Ann. Mean (%)	6.74	3.99	4.36	3.13
Ann. Median (%)	3.10	1.26	2.84	2.71
Ann. St. Dev. (%)	10.00	10.00	10.00	10.00
Skewness	0.37	0.34	0.13	0.55
Kurtosis	3.32	3.71	3.34	3.87
Sharpe	0.67	0.40	0.44	0.31
Sortino	1.31	0.74	0.80	0.62
Max DD (%)	-13.67	-21.62	-14.70	-28.94
Period	01/'92-09/'15	01/'92-09/'15	01/'00-09/'15	01/'00-09/'15

	BarclayHedge		BarclayHedge	
	Systematic TF		Systematic TF	
	AUM	ERW	AUM	ERW
Min (%)	-7.13	-6.13	-7.13	-6.13
Max (%)	10.41	9.94	10.41	9.94
Ann. Mean (%)	7.07	9.25	7.07	9.25
Ann. Median (%)	6.46	7.13	6.46	7.13
Ann. St. Dev. (%)	10.00	10.00	10.00	10.00
Skewness	0.33	0.29	0.33	0.29
Kurtosis	3.28	3.01	3.28	3.01
Sharpe	0.71	0.93	0.71	0.93
Sortino	1.42	1.88	1.42	1.88
Max DD (%)	-11.61	-14.06	-11.61	-14.06
Period	01/'92-09/'15	01/'92-09/'15	01/'94-09/'15	01/'94-09/'15

This Table reports summary statistics for gross and net of transaction costs of ATSMOM in Panel A. Panel B reports the summary statistics for the TSMOM factor of Moskowitz, Ooi, and Pedersen (2012) and the factors proposed by Baltas and Kosowski (2013). Panel C reports the summary statistics for a number of industry benchmarks. The BarclayHedge indices are based on the sample of systematic trend followers selected in Section 4. All time series in the table have been adjusted to 10% annualized volatility for comparison purposes.

Table 4: ATSMOM Strategy Against Existing Risk Factors

Panel A: Gross of Transaction Costs			
	(1)	(2)	(3)
MKT	-0.0565 (0.066)	-0.0691 (0.063)	
SMB	-0.0045 (0.055)		
HML	0.0350 (0.070)		
MOM	0.1964** (0.043)		
LIQ	-0.0865 (0.056)		
PTFSBD		0.0010 (0.014)	
PTFSFX		0.0288* (0.013)	
PTFSCOM		0.0441** (0.015)	
PTFSIR		0.0008 (0.012)	
PTFSSTK		0.0408* (0.018)	
EM		0.0495 (0.043)	
BOND		-0.0157 (0.012)	
CREDIT		1.2142 (1.460)	
GVAL			0.3489* (0.137)
GMOM			0.8587** (0.123)
Ann. Alpha	0.1188** (-0.024)	0.1512** (-0.024)	0.0744** (-0.024)
R-squared	0.129	0.237	0.239
Panel B: Net of Transaction Costs			
Ann. Alpha	0.1016** (-0.024)	0.1335** (-0.024)	0.0579** (-0.024)
R-squared	0.125	0.236	0.236

Panel A reports results for strategy returns gross of transaction costs. Panel B reports the alpha for the same regressions using net-of-transaction costs returns. The factors for the [Fama and French \(1993\)](#) and [Carhart \(1997\)](#) models have been downloaded from Kenneth French's website. The [Pástor and Stambaugh \(2003\)](#) liquidity factor has been obtained from Lubos Pastor's website. The [Fung and Hsieh \(2001\)](#) factors have been taken from David A. Hsieh's Hedge Fund Data Library. The GVAL and GMOM are from AQR's website. * and ** denote significance at the 95 and 99% level, respectively. Robust standard errors in parentheses.

Table 5: Explanatory Power Market-based Indices

Panel A: Adjusted R-squareds						
	Barclay	BTOP50	SG CTA	SG Trend	Barclay Systematic TF (AUM)	Barclay Systematic TF (ERW)
PTFS	0.294	0.190	0.199	0.153	0.218	0.303
F&H 10-Factor	0.329	0.251	0.246	0.198	0.266	0.344
Trend Indicator	0.492	0.482	0.540	0.522	0.507	0.549
MOP	0.431	0.430	0.421	0.459	0.484	0.458
BK	0.413	0.399	0.396	0.419	0.489	0.490
ATSMOM	0.592	0.595	0.604	0.612	0.649	0.658

Panel B: Value Added Beyond Baltas and Kosowski (2013)						
	Barclay	BTOP50	SG CTA	SG Trend	Barclay Systematic TF (AUM)	Barclay Systematic TF (ERW)
BK_M	0.2692** (0.045)	0.4169** (0.053)	0.4028** (0.052)	0.4680** (0.055)	0.5580** (0.057)	0.2717** (0.039)
BK_W	0.2025** (0.061)	0.2840** (0.076)	0.3033** (0.077)	0.3015** (0.082)	0.3340** (0.091)	0.2090** (0.051)
BK_D	0.0895 (0.049)	0.1009 (0.063)	0.0837 (0.068)	0.0723 (0.070)	0.1979** (0.073)	0.1055** (0.038)
Resid	0.2966** (0.052)	0.4584** (0.079)	0.4756** (0.075)	0.4771** (0.083)	0.4969** (0.092)	0.2866** (0.044)
Constant	-0.0006 (0.001)	-0.0040* (0.002)	-0.0037** (0.002)	-0.0045* (0.002)	-0.0023 (0.002)	-0.0003 (0.001)
adjusted R-squared	0.593	0.596	0.607	0.613	0.652	0.676

Panel A reports the adjusted-R²s for regressions where we regress CTA manager-based indices against the asset-based factors. In particular, we consider the Primitive Trend-following Factors (PTFS) of [Fung and Hsieh \(2001\)](#), the Fung and Hsieh's 8-factor model including the remaining two PTFS, the SG trend indicator, [Moskowitz, Ooi, and Pedersen \(2012\)](#) their aggregate time series momentum factor, the Futures-based Trend-following Benchmarks Strategies of [Baltas and Kosowski \(2013\)](#), and ATSMOM. Panel B analyzes the value of including ATSMOM in the model proposed by [Baltas and Kosowski \(2013\)](#). The sample period covered in both panels equals 31/01/2000 through 30/01/2012. * and ** denote significance at the 95 and 99% level, respectively. Robust standard errors in parentheses.

Table 6: Asset Pricing Regressions on Manager-based Indices

	Barclay	BTOP 50	SG CTA	SG Trend	BarclayHedge Syst Trend (AUM)	BarclayHedge Syst Trend (ERW)
Panel A						
ATSMOM	0.4518** (0.047)	0.6332** (0.053)	0.6881** (0.064)	0.6840** (0.071)	0.8200** (0.080)	0.4361** (0.036)
Speed Factor	-0.1473** (0.037)	-0.1219** (0.044)	0.0056 (0.052)	-0.0031 (0.049)	-0.1873** (0.062)	-0.1168** (0.027)
Constant	0.0007 (0.001)	-0.0018 (0.001)	-0.0019 (0.001)	-0.0021 (0.001)	0.0008 (0.001)	0.0014* (0.001)
R-squared	0.528	0.530	0.611	0.605	0.594	0.660
Panel B						
$ATSMOM_{COM}$	0.5617** (0.109)	0.5490** (0.113)	0.6454** (0.142)	0.6479** (0.144)	0.6837** (0.151)	0.5016** (0.072)
$ATSMOM_{EQ}$	0.2544** (0.068)	0.3936** (0.081)	0.4770** (0.091)	0.5898** (0.089)	0.6848** (0.095)	0.2952** (0.047)
$ATSMOM_{FI}$	0.6630** (0.090)	1.1086** (0.126)	1.2303** (0.152)	1.2743** (0.154)	1.4943** (0.140)	0.6705** (0.064)
$ATSMOM_{FX}$	0.4739* (0.215)	0.7027* (0.292)	0.6464* (0.298)	0.4394 (0.274)	0.6500 (0.360)	0.4030* (0.179)
Speed Factor	-0.1370** (0.036)	-0.1154** (0.038)	-0.0008 (0.046)	-0.0074 (0.045)	-0.1617** (0.056)	-0.1062** (0.025)
Constant	0.0005 (0.001)	-0.0021* (0.001)	-0.0021 (0.001)	-0.0023* (0.001)	0.0005 (0.001)	0.0012* (0.001)
R-squared	0.564	0.591	0.676	0.667	0.653	0.697

This table shows the results of the asset pricing regressions against net returns (net of transaction costs) of ATSMOM and the speed factor and the asset class based ATSMOM strategy and speed factor in Panel A and B, respectively. The asset class-based factors are adjusted to 10% annualized volatility. The dependent variables of the regressions are returns of various manager-based indices which are net of transaction costs. * and ** denote significance at the 95 and 99% level, respectively. Robust standard errors in parentheses.

Table 7: Asset pricing regressions on individual Trend-Following Managed Futures

	Ann. Alpha (%)	ATSMOM COM	ATSMOM EQ	ATSMOM FI	ATSMOM FX	Speed Factor	Adj. R ²	Wald
Mean	0.29	0.20	0.12	0.21	0.21	-0.14	0.40	0.00
Mean* (-)	-9.55	-0.37	-0.36	-0.30	-0.18	-0.34		
Mean* (+)	4.77	0.32	0.27	0.31	0.38	0.19		
Median	0.82	0.19	0.13	0.22	0.15	-0.10	0.41	
Median* (-)	-6.56	-0.35	-0.24	-0.25	-0.12	-0.26		
Median* (+)	3.91	0.29	0.23	0.28	0.29	0.15		
t-Statistic	0.22	0.64	0.53	0.70	0.48	0.50		
t-Statistic (-)	0.06	0.03	0.04	0.02	0.03	0.42		
t-Statistic (+)	0.16	0.60	0.49	0.68	0.46	0.08		

This Table shows the mean and median parameter estimates for funds that have at least one-year return history in the dataset and produce jointly significant betas at the 10% level according to the Wald test. The selection procedure leaves us with 335 funds. The rows marked with *(-) and *(+) are in reference to significant negative and positive estimates, respectively. The row named 't-statistic' shows the share of funds that produce significant parameter estimates at 10% significance level. Similarly, the rows labeled 't-statistic (-)' and 't-statistic (+)' show the percentage of funds that produce, at 10% level, significant negative and positive parameters, respectively. The asset class-based ATSMOM and the speed factor are adjusted to 10% annualized volatility and net of transaction costs. The individual fund returns are net of transaction costs and gross of fees. Fees are added assuming a 2/20 fee structure with quarterly crystallization (see Elaut, Frömmel, and Sjödin, 2015).

Table 8: Panel regressions on alphas

	(1)	(2)	(3)
α_{t-1}	0.167** (0.018)	0.200** (0.020)	0.205** (0.022)
Log (FuM)	-0.069* (0.039)	-0.121** (0.042)	-0.145** (0.050)
Age	0.100** (0.019)	0.094** (0.020)	0.093** (0.023)
Fund Flow	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
R ² Perfor. Regr.	3.526** (0.482)	3.500** (0.484)	4.256** (0.532)
Com. Exp.	-1.110 (0.685)	-0.309 (0.713)	-0.168 (0.809)
Eq. Exp.	3.208** (0.747)	3.114** (0.793)	2.679** (0.895)
FI Exp.	-1.472* (0.678)	-1.455* (0.705)	-1.307 (0.783)
FX Exp.	1.110 (0.725)	-1.072 (0.760)	-0.148 (0.852)
Speed Exp.	3.571** (0.626)	3.694** (0.674)	3.467** (0.745)
Mgmt. Fee	-0.251 (0.178)	-0.337 (0.194)	-0.292 (0.226)
Incent. Fee	-0.092** (0.031)	-0.118** (0.035)	-0.162** (0.039)
ME Ratio		0.106** (0.023)	0.128** (0.026)
Round Turns / MUSD			0.000 (0.000)
Observations	2254	2007	1615
R-squared	0.30	0.33	0.37

This Table shows the cross-sectional analysis of the estimated alphas for 335 individual CTAs. The round turns per million USD per year and the margin-to-equity (ME ratio) statistics are not available for each CTA, therefore, in column (2) and (3) we repeat the regressions for the subset of funds for which data are available. The reported coefficients rely on a weighted least squares (WLS) panel regression that accounts for CTA period specific fixed effects. The dependent variable which is the alpha estimates from the performance regressions (see Table 7) is subjected to measurement errors proportional to the standard errors of the performance regressions. Therefore, in the estimation the weights are estimated standard errors of the performance regressions. The standard errors are clustered on both the specific manager and period. * and ** denote significance at the 95 and 99% level, respectively. Robust standard errors in parentheses.

Figure 1: Signal Strength & S&P 500

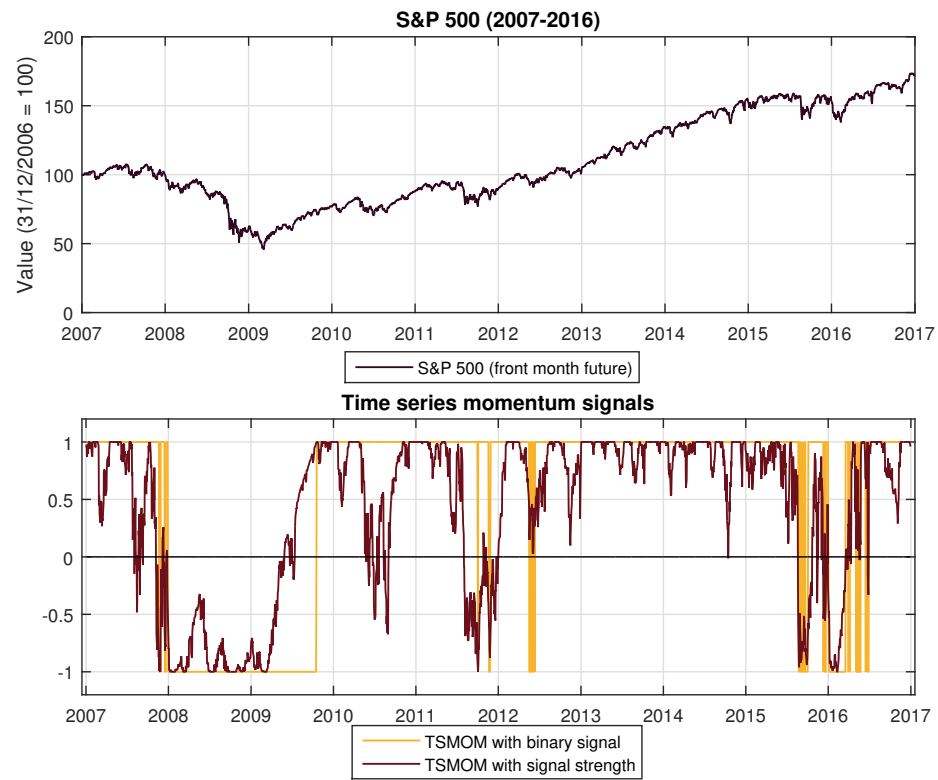


Figure 2: 3-Year Rolling Sharpe Ratio Of Rival Objective Benchmarks

